PYTHON MINI PROJECT

Typing Speed Tester

A desktop application to measure typing speed and character accuracy.



PROJECT TEAM

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Concepts: Strings • Functions • GUI

1. Project Overview

What does it do?

The program shows a sentence, lets the user type it, and reports typing speed, accuracy, and time taken.

Read

A sentence is selected from a small list.

Type

The timer begins with the first key press.

Check

The user finishes the test to see the results.

Purpose: practise basic Python programming by building a useful desktop tool.

2. Features of the Application

Random sentence

Chooses a practice sentence for each round.

Timer

Records the time taken to complete the test.

WPM score

Calculates words typed per minute.

Accuracy

Compares typed characters with the target text.

New Text button

Clears the input and loads another sentence.

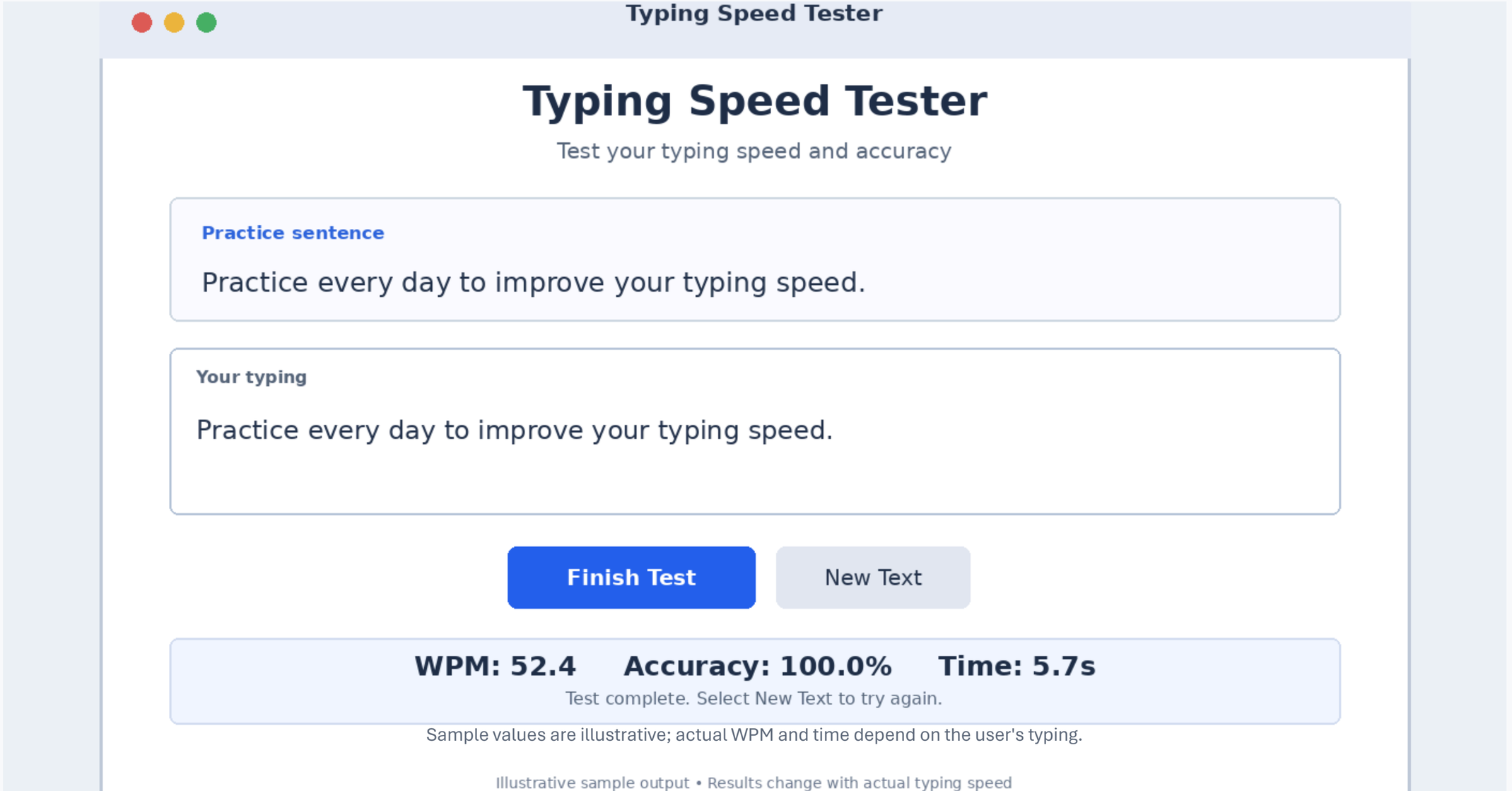
Tkinter interface

Uses labels, a text box, buttons, and result messages.

The project runs with Python's standard Tkinter library; no extra packages are needed.

3. Sample Output

This is an example of what the user sees after finishing a typing test.



4. Python Concepts Used

Strings

- Store the target sentence
- Read typed text from the GUI
- split() counts the words

Functions

- new_test() resets the round
- start_timer() starts timing
- calculate_accuracy() scores text
- finish_test() shows results

GUI (Tkinter)

- Label displays text and results
- Text accepts typing input
- Button triggers actions
- Key events start the timer

Other basics: loops, conditions, random.choice(), classes, and time calculations.

5. Program Workflow

1

Start

Open the app window.

2

Select

Choose a random sentence.

3

Type

First key press starts the timer.

4

Finish

Read the text entered by the user.

5

Calculate

Compute WPM, accuracy, and elapsed time.

6

Display

Update the result label.

The New Text button resets the test so the user can practise again.

6. How the Scores Work

Words Per Minute (WPM)

WPM = words typed ÷ elapsed time in minutes

Accuracy

Accuracy = (correct characters ÷ typed characters) × 100

Example: speed

20 words in 1 minute = 20 WPM

Example: accuracy

45 correct characters out of 50 = 90%

Accuracy is position-based: missing, extra, or misplaced characters can lower the result.

7. Running the Project & Conclusion

How to run

1. Install Python 3.8 or newer.
2. Open the project folder.
3. Run: python main.py
4. Type the sentence and select Finish Test.

What we learned

- Working with strings and functions
- Building a GUI with Tkinter
- Handling button and keyboard events
- Calculating speed and accuracy

Thank you | Questions?